

Validation Engine v2.0

Purpose

The Validation Engine is responsible for comparing information extracted from alcohol label images against application records and generating explainable validation results.

The Validation Engine serves as a decision-support component and does not make final compliance determinations.

All final compliance decisions remain the responsibility of the reviewing compliance agent.

Validation Pipeline

Processing Pipeline

Application CSV

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Label Image

↓

OCR

↓

Text Normalization

↓

Field Extraction

↓

Record Matching



Field Validation



Recommendation Engine



Results Dashboard

The pipeline is intentionally designed to favor transparency and explainability over complexity.

Module Overview

OCR Module

Purpose:

Extract readable text from uploaded label images.

Input:

- PNG
- JPG
- JPEG

Output:

- Raw OCR text

Example:

OLD TOM DISTILLERY

Kentucky Straight Bourbon Whiskey

45% Alc./Vol. (90 Proof)

750 mL

GOVERNMENT WARNING:

...

Text Normalization Module

Purpose:

Convert OCR output into a more consistent format before matching and validation.

Examples:

Example 1

Input:

STONE'S THROW

Output:

stone's throw

Example 2

Input:

750ML

Output:

750 ml

Example 3

Input:

45 % Alc./Vol.

Output:

45% alc./vol.

Benefits

- Improves OCR reliability
 - Reduces false failures
 - Produces more consistent validation behavior
 - Improves record matching accuracy
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Record Matching Engine

Purpose

Identify the most likely application record for each uploaded label.

The Record Matching Engine compares OCR-extracted information against application records loaded from CSV.

Matching factors include:

- Brand Name
- Class / Type
- Alcohol Content
- Net Contents

The engine intentionally uses a conservative strategy to avoid incorrect automatic matches.

Match Confidence Thresholds

Auto Match

Confidence:

95% and above

Action:

Automatically associate the label with the application record.

Needs Review

Confidence:

70%–94%

Action:

Suggest the likely record.

Human confirmation recommended.

No Match

Confidence:

Below 70%

Action:

No automatic association.

Manual review required.

Human Authority

Record matching recommendations are advisory only.

Final record assignment remains the responsibility of the reviewing compliance agent.

Field Extraction Module

Purpose

Identify probable compliance-related values within OCR text.

Core Validation Fields

The prototype extracts:

- Brand Name
- Class / Type
- Alcohol Content
- Net Contents
- Government Warning

These represent the primary validation scope.

Optional Fields

The prototype may also extract:

- Bottler / Producer
- Country of Origin
- Beverage Type

These fields are included for future extensibility.

Validation Categories

Exact Match

Used when values must be legally or numerically precise.

Fields:

- Alcohol Content
- Net Contents

Example:

Expected:

45%

Detected:

40%

Result:

Fail

Strict Match

Used when regulatory language must remain exact.

Fields:

- Government Warning

Purpose:

Federal regulations require specific warning language.

Example:

Expected:

GOVERNMENT WARNING:

Detected:

Government Warning:

Result:

Fail

Fuzzy Match

Used when formatting differences should not trigger failures.

Fields:

- Brand Name
- Class / Type

Purpose:

Handle:

- Capitalization differences
- Minor formatting differences
- OCR formatting inconsistencies

Example:

Expected:

STONE'S THROW

Detected:

Stone's Throw

Result:

Pass

Normalized Match

Used when formatting differs but meaning remains identical.

Example:

Expected:

750 mL

Detected:

750ml

Result:

Pass

Normalization may remove:

- Extra spaces
- Case differences
- Limited punctuation differences

Normalization does not ignore actual value differences.

Field Status Categories

Pass

The detected field matches the expected application value.

Fail

The detected field clearly conflicts with the expected application value.

Missing

The required field was not found.

Examples:

- Missing alcohol content
 - Missing government warning
 - Missing net contents
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Needs Review

The system cannot confidently determine the result.

Examples:

- OCR uncertainty
- Poor image quality
- Ambiguous record match
- Partial extraction
- Conflicting indicators

The prototype intentionally favors Needs Review over incorrect pass/fail decisions.

OCR Failure Handling

Purpose

Prevent OCR quality problems from generating incorrect compliance outcomes.

Rule

If OCR cannot confidently extract sufficient information:

Result:

Needs Review

Reason:

Poor image quality should not automatically fail a label.

Examples

Blurry Label

Result:

Needs Review

Severe Glare

Result:

Needs Review

Cropped Label

Result:

Needs Review

Recommendation Engine

Purpose

Convert field-level validation results into an overall recommendation.

The Recommendation Engine is intentionally conservative.

Possible Outputs

PASS

All required fields validated successfully.

No significant mismatches detected.

FAIL

One or more required fields clearly conflict with expected application values.

Examples:

- Incorrect ABV
 - Incorrect net contents
 - Incorrect government warning
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NEEDS REVIEW

One or more fields are:

- Missing
- Low confidence
- Ambiguous
- OCR uncertain
- Poor quality

Human review recommended.

Overall Status Logic

PASS

Requirements:

- All required fields pass
 - Record match confidence acceptable
 - No unresolved issues
-

FAIL

Requirements:

- One or more required fields clearly fail validation
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NEEDS REVIEW

Requirements:

- OCR uncertainty
- Missing fields
- Ambiguous record match
- Confidence concerns

The application should favor Needs Review whenever uncertainty exists.

ERROR

Requirements:

- Unsupported file
- Corrupt image
- Processing exception

Error indicates a system problem rather than a compliance outcome.

Explainability Requirement

Every result generated by the Validation Engine must include:

- Expected Value
 - Detected Value
 - Match Method
 - Confidence Score
 - Explanation
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Example Result

Field:

Brand Name

Expected:

STONE'S THROW

Detected:

Stone's Throw

Method:

Fuzzy Match

Confidence:

100%

Explanation:

Brand name matches after normalization.

Future Validation Expansion

Future production versions may add:

Beverage-Specific Validation

- Beer validation rules
- Wine validation rules
- Distilled spirits validation rules

Layout Validation

- Government warning placement
 - Required section positioning
 - Label structure checks
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Font Validation

- Minimum font size
 - Font weight requirements
 - Visibility checks
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Advanced OCR

- Region detection
- Label segmentation
- Orientation correction
- Enhanced image preprocessing

These capabilities are intentionally excluded from the prototype scope to maintain focus on the core stakeholder requirements and delivery timeline.

Design Philosophy

The Validation Engine prioritizes:

- Explainability
- Transparency
- Consistency
- Human review
- Conservative decision-making

The objective is not to replace compliance agents.

The objective is to reduce repetitive verification work while preserving accountability and regulatory judgment.